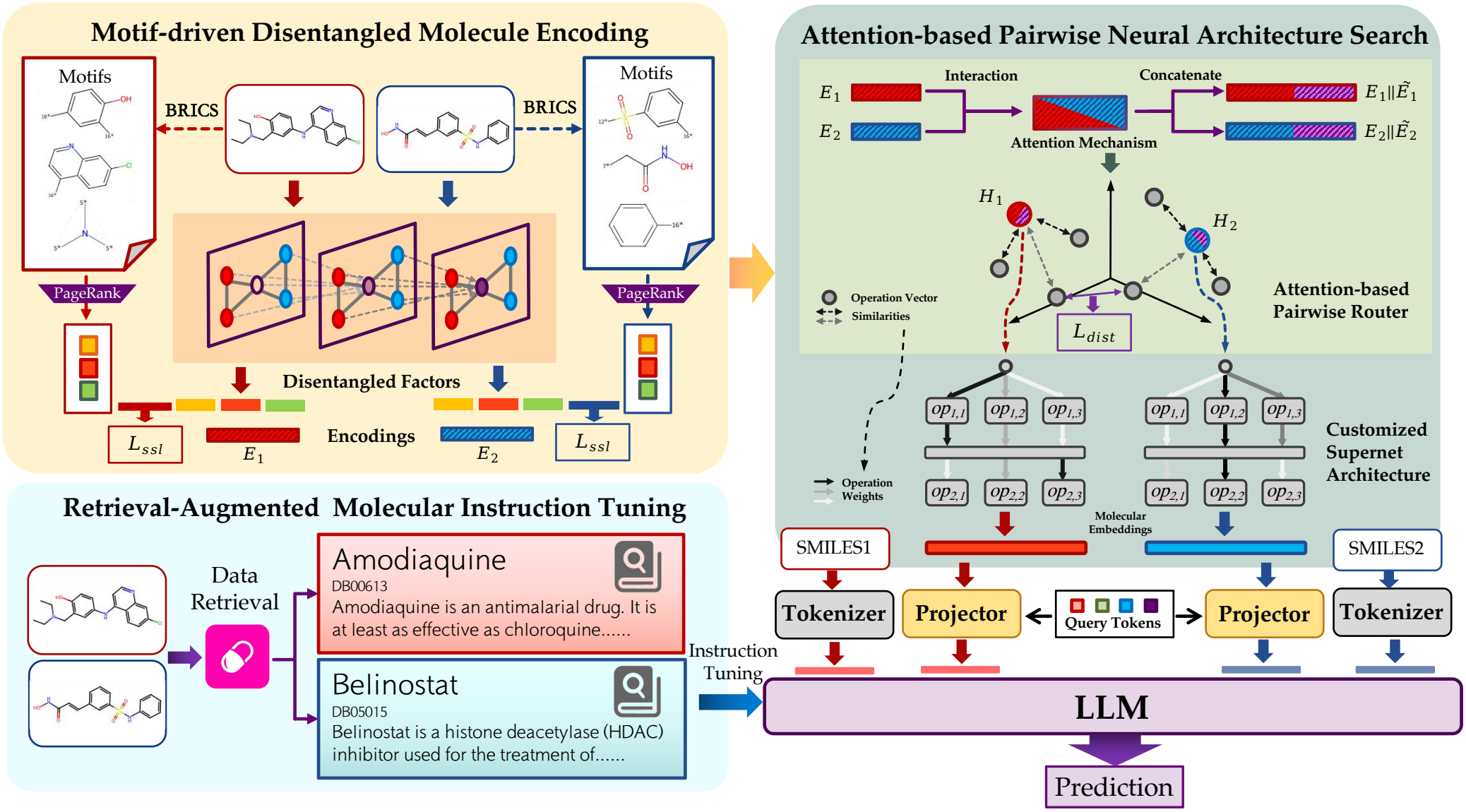

Motif-driven Disentangled Molecule Encoding
Motifs
BRICS
BRICS
Motifs
PageRank
PageRank
Disentangled Factors
L_{ssl}
E_1
Encodings
E_2
L_{ssl}
Attention-based Pairwise Neural Architecture Search
E_1
E_2
Interaction
Attention Mechanism
Concatenate
$E_1||\tilde{E}_1$
$E_2||\tilde{E}_2$
H_1
H_2
Operation Vector Similarities
L_{dist}
Attention-based Pairwise Router
Operation Weights
$op_{1,1}$ $op_{1,2}$ $op_{1,3}$
$op_{2,1}$ $op_{2,2}$ $op_{2,3}$
Customized Supernet Architecture
Molecular Embeddings
SMILES1
SMILES2
Tokenizer
Projector
Query Tokens
Projector
Tokenizer
Retrieval-Augmented Molecular Instruction Tuning
Data Retrieval
Amodiaquine
DB00613
Amodiaquine is an antimalarial drug. It is at least as effective as chloroquine.....
Belinostat
DB05015
Belinostat is a histone deacetylase (HDAC) inhibitor used for the treatment of.....
Instruction Tuning
LLM
Prediction